Book strip

Science · LAUNCH · Autumn 2 · Week 2 · SCI-LAUNCH-A2-W02-L2

Use: how stem cells become specialised cells

I can explain how a stem cell divides and how some of its descendants become specialised for a job.

Success: Tell division from differentiation. · Link a specialised cell feature to its job. · Compare an animal stem-cell route with growth at a plant meristem.

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Route card

Staff: cut these apart before the lesson. Give one route card. Keep the others as alternatives. Record actual I · S · C beside each task.

Supported



Supported
Feature → function → reason

Supported route card: connect a cell feature to its function

Finish: Division makes more ___. Differentiation gives a specialised ___. [2]

Use Reference A and B. Complete the matching task sheet in your book or usual mode.

Staff I ☐ S ☐ C ☐

Independent



Independent
Feature → function → reason

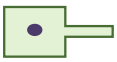
Independent route card: connect a cell feature to its function

Explain the difference between division and differentiation. [2]

Use Reference A and B. Complete the matching task sheet in your book or usual mode.

Staff I ☐ S ☐ C ☐

Aspire



Aspire
Feature → function → reason

Aspire route card: connect a cell feature to its function

A model shows two new cells, both labelled specialised. Explain why cell number alone cannot justify those labels. Add the missing process. [3]

Use Reference A and B. Complete the matching task sheet in your book or usual mode.

Staff I ☐ S ☐ C ☐

Choice



Choice
Feature → function → reason

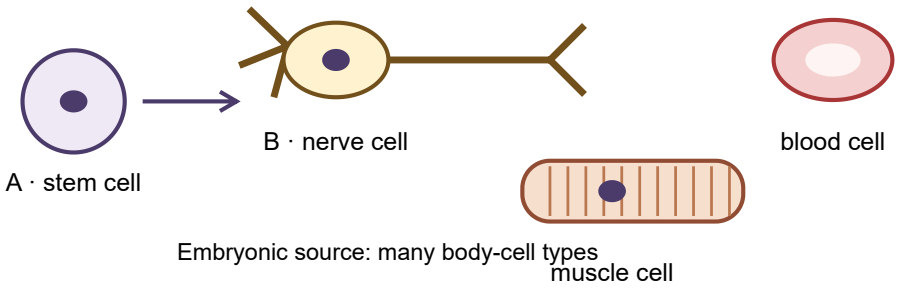
Choice route card: connect a cell feature to its function

Choose division or differentiation for makes more cells. Choose again for gives a job. Add one short reason. [2]

Use Reference A and B. Complete the matching task sheet in your book or usual mode.

Staff I ☐ S ☐ C ☐

Reference A animal cells



An embryonic stem cell can give rise to many specialised body-cell types.

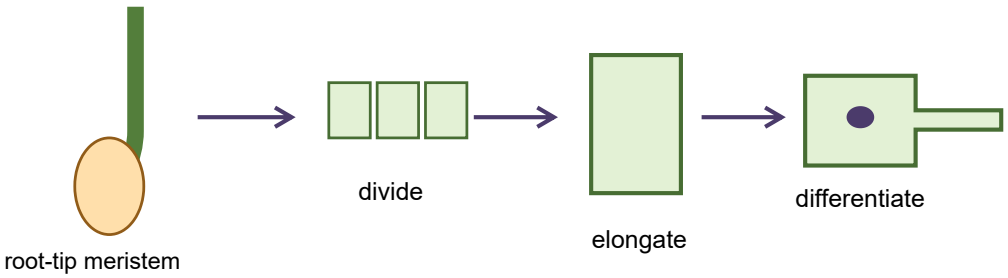
Reference only. Do not write on this sheet. Trace a feature, then explain its function.

Cell or source	Feature or range	Function or limit
Nerve cell	Long extension	Carries signals over a distance
Muscle cell	Structures that contract	Contraction produces force
Embryonic stem cell	Can give rise to many body-cell types	Not a promise of a successful treatment
Adult blood-forming stem cell	More limited range of blood-cell types	Makes replacement blood cells

The arrows show possible lineages. The drawing does not show every cell stage. Cells do not choose their own jobs. Specialised cells can share inherited DNA but use it differently.

Made by Matt

Reference B root growth



Root-tip meristem, division, elongation and differentiation in a simplified route.

Reference only. Do not write on this sheet.

Process or region	What it adds	Useful limit
Meristem	A region where cells divide	Not every root cell remains a meristem cell
Division	More cells	Number alone does not prove a specialised role
Elongation	Longer cells	Not the name for becoming specialised
Differentiation	Features suited to a job	Different cells can be at different stages

A root hair extension provides a large surface for absorption. The route is simplified. Use it to compare processes, not as a claim that every cell follows an identical sequence.

Made by Matt

Catch-up sheet absence or access only

Use only after absence or when an access barrier prevents joining the current task. It replaces a starting step; it adds no missed-work demand.

1. Division makes more cells. Differentiation gives cells specialised roles.

2. A stem cell can divide and give rise to specialised cell types. Adult stem-cell types have more limited ranges than embryonic stem cells.

3. In a root-tip meristem, cells divide. New cells can elongate and differentiate.

Show one link: more cells → division; a job → differentiation.

Then join the current model or You do task 1. Use Reference A or B beside the task.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

Made by Matt

Choice task sheet

Choose a response, then give a short reason. Use Reference A and B.

1. Choose division or differentiation for makes more cells. Choose again for gives a job. Add one short reason. [2] Staff I ☐ S ☐ C ☐

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2. Choose nerve or muscle for carrying signals. Choose long extension or no structure as a useful feature. Explain one link. [3] Staff I ☐ S ☐ C ☐

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3. Choose blood cells or all body cells for the range shown from an adult blood-forming stem cell. Say one limit. [3] Staff I ☐ S ☐ C ☐

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4. Select a plant sequence: divide then elongate then differentiate, or differentiate then no further change. Name root-tip meristem or leaf middle as the dividing region. [4] Staff I ☐ S ☐ C ☐

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5. Select both, animal only or plant only for division. Select elongation or signal transmission as the length change in the root model. Give one comparison sentence. [3] Staff I ☐ S ☐ C ☐
